

SYLARNA, Sweden

WMO: 022090

Lat: 63.0438N

Lon: 12.2748E

Elev: 3379

StdP: 12.99

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.9	-1.7	-18.6	2.2	-3.9	-10.8	3.5	2.3	51.8	20.0	47.4	20.7	7.9	160	0.948

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	66.1	53.9	62.7	52.8	59.7	51.3	56.5	63.4	54.5	60.7	52.5	58.2	10.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.4	68.7	59.1	51.1	63.1	57.3	49.3	59.0	55.8	25.5	63.5	24.1	60.7	22.9	58.3	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.4	36.7	32.3	DB	-7.4	69.9	10.3	3.6	-14.8	72.5	-20.8	74.6	-26.6	76.6	-34.1	79.3	(4)
(5)			WB	-7.1	58.8	9.6	3.1	-14.1	61.0	-19.7	62.8	-25.1	64.5	-32.1	66.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.5	19.7	18.0	21.0	29.1	36.4	43.9	49.5	48.3	41.9	32.3	25.9	22.7		
	DBStd	13.41	8.72	10.57	7.98	7.05	7.56	7.09	6.60	6.36	6.39	7.07	8.68	8.64		
	HDD50	6572	939	895	900	628	429	209	90	109	253	549	724	847		
	HDD65	11874	1404	1315	1365	1078	888	634	480	517	693	1014	1174	1312		
	CDD50	173	0	0	0	0	6	25	75	57	10	0	0	0		
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH74	3	0	0	0	0	0	0	2	1	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.8	18.3	17.6	16.7	14.3	12.8	12.2	10.9	11.6	14.0	15.5	16.4	17.8		
	PrecAvg	31.8	2.6	2.2	2.2	1.5	1.9	3.2	4.0	3.7	3.3	2.7	2.3	2.2		
(15) Precipitation	PrecMax	41.1	5.9	5.4	4.6	4.0	3.3	4.8	7.1	5.9	7.4	4.6	6.2	5.0		
	PrecMin	22.1	0.1	0.1	0.8	0.4	1.0	1.3	1.3	1.4	1.1	0.9	0.1	0.5		
	PrecStd	4.0	1.6	1.1	1.1	0.9	0.6	1.0	1.5	1.4	1.7	0.9	1.7	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	42.2	40.8	47.5	63.0	66.8	72.3	68.8	60.7	52.9	46.8	42.7		
		MCWB	34.3	35.4	34.0	40.7	51.2	53.8	57.1	55.1	51.0	47.3	39.2	36.4		
	2%	DB	35.2	36.3	36.9	44.5	56.1	62.3	67.5	64.5	56.8	46.7	42.8	39.1		
		MCWB	31.4	31.4	32.8	38.6	46.8	50.9	54.7	53.8	49.4	42.0	38.3	34.7		
	5%	DB	32.5	32.5	34.0	41.5	51.3	58.3	63.6	61.3	54.0	44.0	39.2	36.4		
		MCWB	29.6	29.8	31.2	36.4	43.8	49.1	53.4	52.7	48.2	40.0	36.2	32.8		
	10%	DB	30.3	29.7	31.4	38.8	47.4	55.0	60.3	58.1	51.3	41.8	36.3	33.2		
		MCWB	28.0	27.4	29.0	34.6	41.3	47.6	51.8	51.2	46.6	38.2	33.5	30.3		
	0.4%	WB	34.6	36.6	35.9	42.6	52.4	56.5	59.3	58.4	53.3	48.8	41.9	38.7		
		MCDB	37.9	40.8	38.4	45.9	60.2	61.9	68.1	64.6	57.3	52.0	44.9	41.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.5	32.3	33.6	39.3	47.4	52.3	57.1	55.8	51.0	42.9	39.0	35.5		
		MCDB	34.9	34.7	35.9	43.5	54.6	59.1	65.3	61.7	54.9	45.6	41.7	38.2		
	5%	WB	30.5	29.8	31.2	36.9	44.4	50.2	55.2	53.9	48.9	40.7	36.3	32.8		
		MCDB	32.4	32.0	33.6	40.7	50.2	57.1	62.1	59.7	52.9	43.7	38.8	35.7		
	10%	WB	28.2	27.5	29.1	35.1	41.8	48.1	53.1	52.0	46.7	38.7	33.6	30.6		
		MCDB	29.8	29.4	31.2	38.3	47.1	54.5	59.2	57.3	50.5	41.3	36.2	32.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.1	8.9	8.3	9.9	11.2	11.1	9.7	8.8	7.4	8.2	8.6		
		MCDBR	9.8	11.2	10.3	9.7	14.8	16.7	16.8	14.1	12.6	10.2	9.3	11.4		
		MCWBR	8.5	8.4	8.4	7.1	9.5	10.0	10.0	8.4	8.2	8.2	7.9	9.3		
		MCDBR	9.5	10.4	9.8	9.2	14.0	15.0	14.7	12.6	10.7	10.0	9.5	10.9		
	5% WB	MCWBR	8.7	8.4	8.5	7.1	9.9	10.0	9.9	8.5	8.0	8.4	8.7	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.181	0.239	0.266	0.292	0.304	0.304	0.314	0.304	0.281	0.261	0.190	0.162		
	taud		2.036	2.257	2.269	2.298	2.453	2.538	2.521	2.539	2.590	2.479	2.090	1.955		
	Ebn at Noon		159	233	269	284	290	292	286	280	268	226	151	93		
	Edh at Noon		9	18	26	32	30	29	28	25	20	15	9	5		
(44) All-Sky Solar Radiation	RadAvg		61	240	683	1267	1485	1563	1481	1133	692	300	87	30		
	RadStd		7	35	70	100	104	141	168	111	69	38	13	2		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A	(47)

Nomenclature: See separate page